

E.N.G.R.: Engineering Navigation Guidance Robot

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https://github.com/miltonT2023/857_Final_Project_Code

Abstract—An assistive guidance robot has the potential to improve the navigation experience, reducing stress and improving timeliness for first-time visitors to a building. At San Francisco State University, the current approach relies on hard-to-find maps that are often outdated and lacking in detail or through trial-and-error. In both cases, visitors are uncertain of their arrival and therefore arrive late or need to allocate excessive travel time. Here, we implement a robot that greets potential users, interprets natural language, and accurately navigates to guide users to their desired location. In subject testing, our prototype’s social interaction abilities were well received; 5/5 participants felt comfortable using it. Our other results emphasize the importance of predictability and speed. This work demonstrates the potential of a socially interactive guide robot.

I. INTRODUCTION

In the past decade, robot deployment has doubled [1]. However, this growth is primarily limited to factory settings. This isn’t due to a lack of interest; Zimmerman et. al. [2] reviews over 1400 papers on human-robot-interaction (HRI) published between 2015 and 2021. It’s due to, as they describe it, the interdisciplinary nature of HRI in which technical implementation is only one step in building a robot that can assist humans. This is amplified in non-factory settings where context is less controlled.

Work by Pineda et. al. [3] shows the positive effect of small talk on human-robot collaboration while noting ethical concerns about trust and obligation. In [4], Ono et. al. demonstrates a robot that implicit guides users through use of a projector rather than explicit instructions.

Using these works as a basis, this project explores the development of a human-centered social guidance robot for the Science and Engineering Innovation Center (SEIC) at San Francisco State University. To evaluate the effectiveness of this system, the project focuses on the following research question: How effectively can users interact with a human-centered social guidance robot, understand its communication cues, and complete a room-navigation task within the SEIC building? The proposed system was developed on the Quanser QBot platform using ROS 2 to connect the robot’s perception, interaction, mapping, and navigation modules. The robot uses camera-based person detection and a graphical user interface (GUI) to begin the interaction and receive a destination request from the user. During system setup, SLAM is used to create a map of the environment,

and selected rooms or locations are labeled as destination points. During navigation, Nav2 uses the labeled map to plan a path and guide the robot toward the selected destination. Throughout the interaction, the GUI, speech output, display messages, and LED cues provide feedback so users can better understand the robot’s current state and actions.

II. SYSTEM OVERVIEW

This system was developed using the Quanser QBot Platform. The QBot Platform is an educational mobile robot with built-in sensors and software modules for robotics development. The robot includes several sensors applicable to navigation and perception. For 2D, 360° range-finding, it has the Leishen M10P. For forward-facing vision, the robot also has the Intel RealSense D435 RGB-D camera. In addition, the robot includes wheel encoders, digital tachometers, current sensors, and a 6-axis inertial measurement unit (IMU) for odometry, motor feedback, and acceleration feedback. For running software and system integration, the system uses an NVIDIA Jetson Nano Orin on an AverMedia D131-L carrier board.

To support communication between the user and the robot, external interaction components were added. These included a keyboard for receiving user input, a speaker for verbal instructions, and a low power monitor for displaying the user-facing module. The keyboard and monitor are placed on a 6-legged raised surface mounted onto the QBot for natural interaction.

Our software was developed using ROS2 [5] as our middle-ware suite. Through ROS2, we implemented components of existing libraries and models and wrote custom nodes. We also made several non-user facing GUIs in Python.

A. Robot Control System

The robot behavior is organized into four main stages: waiting, confirmation, navigation, and arrived. These stages allow the robot to manage the interaction flow between the user and the system.

In the waiting stage, the robot remains idle and waits for a user to approach. During this stage, the computer vision system detects the presence of a person. When a person is detected, the robot gives a friendly greeting through speech and displays a greeting message on the GUI. The GUI border also turns green to show that a person has been detected. The

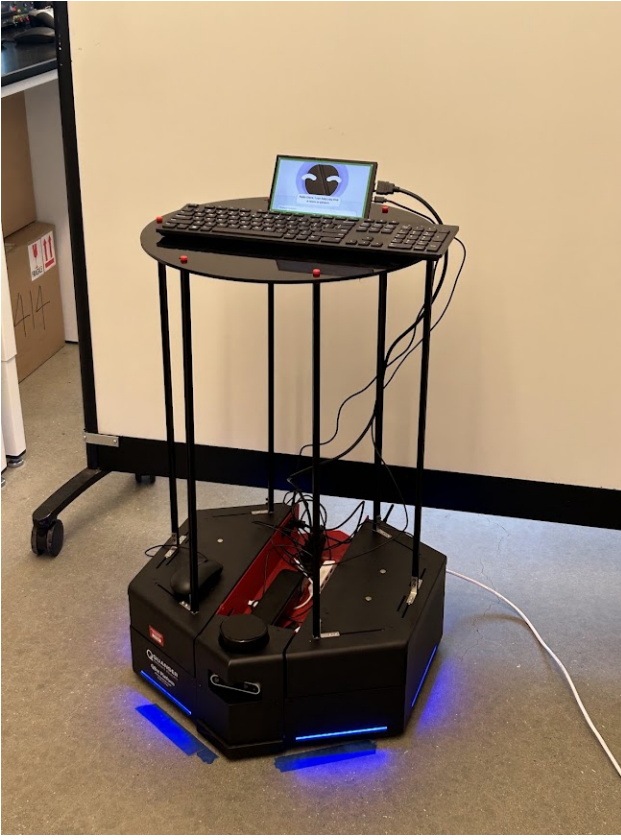


Fig. 1: Picture of the E.N.G.R. The baseline QBot Platform with no additions is the black hexagonal prism on the floor. Aside from the cylindrical protrusion, the LiDAR, all other components above the prism are additions.

robot explains that it can help the user find a room or person in the building. During the waiting stage, the robot LEDs remain blue.

After the user enters the name of a person or room, the system moves into the confirmation stage. In this stage, the robot searches for the requested destination and provides information about the room or person, such as the room location or floor. The robot then asks the user whether they would like to be guided to that destination. During this stage, the LEDs blink green to show that the robot is waiting for confirmation. If the user selects no, the robot returns to the waiting stage and the LEDs turn blue again. If the user selects yes, the robot begins navigation.

In the navigation stage, the robot publishes the selected destination label to the navigation system. The LEDs turn solid green, and the robot announces that it is moving to the target location. The Nav2 system then plans and executes the path to guide the user to the selected room.

When the robot reaches the destination, it enters the arrived stage. In this stage, the robot announces that it has arrived and thanks the user. The system waits 10 seconds before starting the return process. After this pause, the robot navigates back to its original starting position. Once it reaches the origin, the system resets and returns to the

waiting stage.

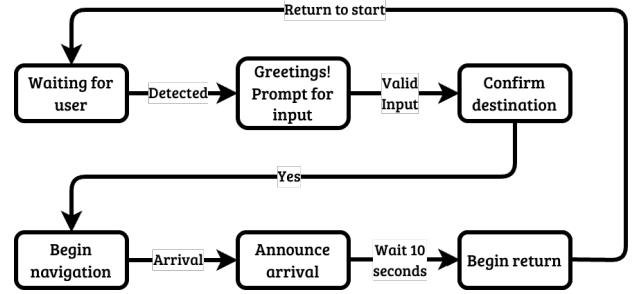


Fig. 2: Simplified behavior flow chart of E.N.G.R. where the user all answers are valid and confirmed.

B. User-Facing Module

The user-facing module provides a simple visual interface for the user. It displays an animated robot face with moving eyes, as shown in Fig. 3, the GUI displays the robot face and user input area.

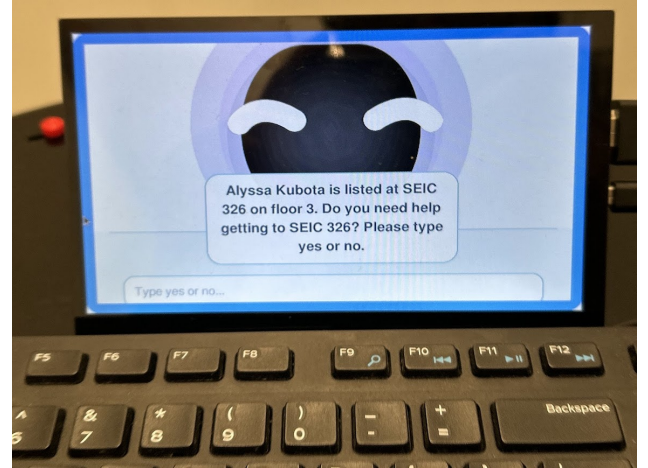


Fig. 3: Picture of the user-facing interface of the guidance robot.

The user-facing module also includes a text area showing messages from the robot such as greetings, destination information, confirmation prompts, and navigation status updates. At the bottom of the GUI, there is an input box where the user can type the name of a room or person. This allows the user to interact with the robot directly and clearly see what they are entering before submitting the request.

C. YOLO Model for Computer Vision Object Detection

For the computer vision object detection module, the system uses the pretrained YOLOv8n model from Ultralytics. YOLOv8n is the nano version of the YOLOv8 model family, making it one of the smallest and most lightweight version. This was suitable for the guidance robot because the system required real-time detection while running on limited onboard computing resources.

In this project, the YOLOv8n model was not trained on a custom dataset. Instead, the pretrained model was

used to detect when a person was standing in front of the robot. Although the model can detect many different object categories, the system only uses the person class and ignores the other detected object labels. This allows the robot to focus only on detecting human presence.

D. Mapping and Navigation System

In order to guide users, our robot requires a map of the space. We used Cartographer to perform simultaneous localization and mapping (SLAM). Our sensor inputs for this are the LiDAR, wheel odometry, and IMU. In order to improve localization and mapping accuracy, we implemented a "wedge" shaped filter GUI, shown in Fig. 4 for the LiDAR output; it's primary purpose was to filter inaccurate readings caused by the 6 poles holding the raised surface up. For mapping, the robot was driven manually by controller. After the map was created, we used another GUI, shown in Fig. 5 with a list of all rooms in the SEIC building for labeling.

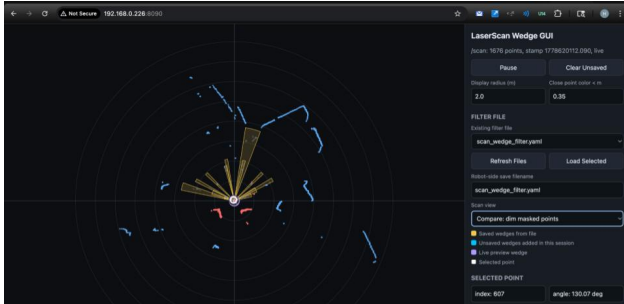


Fig. 4: GUI for filtering LiDAR scans. Left shows live LiDAR scans as circles with wedges filtering out any circles contained. Right shows options for GUI such as pausing and wedge editing. Wedges are saved to a .yaml file to be read by the filter.

During navigation, our robot localizes using the same sensor fusion as during SLAM. For path planning and velocity commands, we use Nav2. Our Nav2 implementation checks the state of the robot and if the HRI component has output a destination label; when the robot enters navigation mode and receives this destination label, Nav2 finds the matching label on the map and begins path planning and driving using a cost map. While on the way, it also performs obstacle avoidance using the filtered LiDAR output. Once it arrives, it notifies the user and after a delay, returns to its starting point.

E. Human-Robot Interaction Features

Using YOLO and the camera's built-in depth functionality, E.N.G.R is able to detect approaching people. Once within range, the robot greets these potential users, beginning interaction.

This design connects to prior work in social robotics, where embodied robots use verbal and nonverbal cues to support human interaction. In Shen et. al [6], the robot uses speech, physical embodiment, and a tablet display to guide users through an interaction. Similarly, our system uses

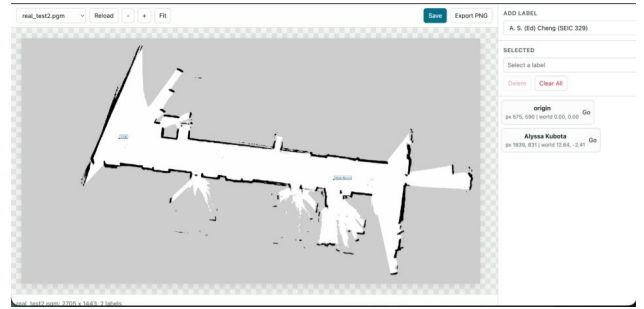


Fig. 5: GUI for displaying map and adding labels. Left shows image of map based on LiDAR scans; right shows dropdown for adding labels, existing labels, and their positions on the map and how that corresponds to the actual physical space.

human detection, speech, and a visual display to create a clear starting point for user interaction.

The robot's greeting is: "Hello, I am a guidance robot. I can help you find a location or person in this building." This greeting lets the user know that the robot is ready to help and to clearly explain the robot's role. The same message is also displayed on the monitor so the user can both hear and read the instructions. This multi-modal approach makes the interaction easier to understand for across users with different preferences and capabilities.

F. Workflow

The workflow of the system is divided into two main parts: the setup workflow and the user interaction workflow. The setup workflow is first completed before the robot can guide users. First, the robot is manually driven through the target area of the SEIC building while Cartographer SLAM is used to create a map of the environment. After the map is created, rooms and important target locations are labeled using the map-labeling GUI. These labels allow the navigation system to connect a user request, such as a room or person, to a physical location on the map.

Once the map and labels are ready, the robot is placed at its starting location. This starting point is treated as the robot's origin. During operation, the system uses the GUI, speech, computer vision, and navigation modules together to support the user interaction. The robot waits for a user, provides instructions, receives a destination request, and uses the labeled map to guide the user to the selected location.

Once the robot reaches the target destination, it provides arrival feedback to the user through speech and the display. After a short delay, the robot returns to its original starting location. When the robot reaches the origin, the system resets and resumes waiting for the next user interaction.

III. METHODOLOGY

A. Technical Evaluation

The system was tested in individual steps by module, followed by a final integrated system test. Each major

component was evaluated separately to verify proper functionality before testing the complete system as a whole.

The user-facing module was tested through interaction-based testing. During this process, we verified that the system progressed step by step through the expected interaction flow. The speech activation phrases were tested to confirm that the correct text responses were displayed in the interface. The LED feedback system was also tested to verify that the LEDs responded correctly during each stage of the interaction.

The YOLO-based person detection system was tested by having a person move in front of the robot. This allowed us to verify that the robot could detect a person and respond appropriately based on the intended behavior. In addition, all ROS topics and subscribers were reviewed to confirm that the correct data was being published, received, and used by the system.

The system was developed and evaluated in a research lab environment. The SLAM system was tested by mapping a selected area of the room and comparing the generated map with the actual physical layout. The accuracy of the map was evaluated by verifying the placement of key room features, labeled locations, and the estimated robot position. This process was repeated from multiple starting locations to assess the consistency of the SLAM output.

After the map was generated and labeled, the Nav2 navigation system was evaluated through multiple navigation trials. In each trial, the robot was commanded to navigate from its initial position to a selected destination within the mapped environment. These tests were used to determine whether the robot could generate a valid path, navigate through the environment, and reach the intended target location.

After verifying the individual modules, the system was tested as a fully integrated platform. A complete interaction sequence was performed in which the user moved through the system phases in the intended order. Additional tests were also performed by triggering certain steps out of order to evaluate how the system responded to unexpected inputs or unplanned sensor activations. Based on these tests, the system was able to return to a safe default state when necessary. This confirmed that the system was stable enough for controlled user testing.

B. Human Evaluation

For the final test, the robot was evaluated in the third-floor hallway of the SEIC building. First, the hallway was mapped using the SLAM system. After the map was created, the robot was placed at its origin point, and the system was turned on in its waiting state.

During the study, participants were given limited information about the robot. This was done to observe whether users could naturally understand how to approach the robot, begin the interaction, and follow its communication cues. The only instruction given was if the robot asked for a destination, the participant should type “Dr. Teh,” a known person in the building, and confirm the destination. Each participant was placed near the robot and told they could begin interacting with it at any time. The participant then approached the robot,

followed the interaction process, entered the destination, and started the navigation task. During the trial, observations were recorded to evaluate how the participant understood the robot’s cues, entered the destination, and followed the robot during navigation. If the robot successfully reached the destination, the participant was given a survey immediately after arrival. The robot then attempted to navigate back to the original starting location. If the robot became stuck or failed to reach the destination, the test was ended and the participant was still given the survey. After each trial, the robot was reset to the origin before the next participant began. Throughout the test, the only information collected was observation notes and survey responses related to each participant’s interaction with the robot. These notes and survey responses are summarized in the following section of this report.

IV. RESULTS

The final evaluation included five participants. The survey responses were summarized into three categories: Yes, Partial/Maybe, and No. Table I shows a summary of the participant survey results. The survey was designed to evaluate both the technical performance of the navigation system and the human experience of interacting with the robot. The questions focus on several key areas, including communication clarity, navigation support, user comfort, safety, trust, and usefulness of the system. During the trials, several important behavioral observations were recorded. One of the most noticeable reactions was participant uncertainty and discomfort in response to the robot’s voice. Some participants described the voice feedback as surprising or unsettling, especially when the robot began speaking unexpectedly. Another common observation was that participants quickly moved away from the robot when it started moving, particularly when they were standing in front of it. Throughout the trials, the robot occasionally stopped and turned side to side. This behavior occurred as part of the robot’s navigation process, where the system used LiDAR data to localize itself, reorient, and determine the correct path. However, during these moments, several participants appeared unsure about what to do next and looked toward the observer for guidance. Since the observer was not allowed to interact with the participant during the trial, this uncertainty remained part of the user experience. When following the robot to the target room, some participants did not walk directly beside or closely behind the robot. Instead, they maintained an observation distance of approximately five feet.

V. DISCUSSION AND FUTURE WORK

From the survey results, participants generally understood the basic interaction flow between the human and the robot. Most users were able to understand when the robot greeted them, asked for a destination, confirmed the destination, and instructed them to follow it. This suggests that the GUI, screen display, and basic communication structure were effective in guiding the user through the interaction.

TABLE I: Summary of Participant Survey Results

Survey Item	Yes	Partial	No
I understood when the robot was greeting me.	4	0	1
I understood when the robot was asking for my destination.	5	0	0
I understood when the robot confirmed the destination.	3	2	0
I understood when the robot wanted me to follow it.	4	0	1
The robot's screen or face display was clear.	4	1	0
The robot's light/LED cues helped me understand its status.	2	1	2
The robot's movement made it clear where I should go.	2	1	2
I knew when I needed to respond to the robot.	2	1	2
The robot helped me reach the correct room/location.	2	0	3
The path taken by the robot was easy to follow.	5	0	0
The robot moved at an appropriate speed.	1	0	4
I did not feel lost while following the robot.	1	0	4

Survey Item	Yes	Partial	No
The robot made the navigation task easier.	2	1	2
The robot's behavior was predictable.	2	1	2
The robot gave me enough information to complete the task.	3	0	2
I would use this robot again to find a room in the SEIC building.	1	1	3
I felt safe walking near the robot.	4	0	1
I trusted the robot's directions.	2	0	3
The robot kept a comfortable distance from me.	3	1	1
I felt comfortable following the robot through the hallway.	3	2	0
I felt the robot understood the task.	4	1	0
I felt comfortable interacting with the robot.	5	0	0
I would recommend this robot to someone unfamiliar with the building.	3	1	1

Another important finding is that the path itself was generally easy to follow, but users still felt uncertain during some parts of the task. This uncertainty may have been caused by short pauses, turning behavior, or the lack of explanation during navigation adjustments. Trust in the robot was mixed, which suggests that the system does not appear reliable from the user's perspective. Most participants reported feeling safe near the robot and comfortable following it through the hallway, but some users still kept distance from the robot while it was moving. In addition, the robot's voice affected user comfort because some participants found it surprising or unsettling.

VI. CONCLUSION

This project developed a human-centered social guidance robot for indoor navigation within the SEIC building. The system combines SLAM, Nav2, a GUI, speech feedback, LED cues, and display messages to help users select a destination and follow the robot to a target room. By integrating autonomous navigation with social interaction cues, the project explores how a guidance robot can support users in navigating an unfamiliar educational environment. The results showed that participants generally understood the basic interaction flow and felt comfortable interacting with the robot. However, the results also showed that the system still needs improvement before it can be considered fully reliable. Future work for the system will include improving navigation reliability, motion predictability, voice input, and clearer status communication so users can better understand what the robot is doing during the task.

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